

Volume-Resolved Phonon Constraints for Linear ACE Committees

Mathematics, algorithm, and the code that implements it

Model: A1_12_4_6A_2_ (full dataset, $K = 91$ parameters)

Generated 27 July 2026 — repository
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Abstract

This document sets out, in a form directly translatable into a paper algorithm, the constrained-committee method used for A1_12_4_6A_2_: how a linear ACE potential’s parameter vector is fitted under physical constraints, how a POPS uncertainty committee is generated from it, and — the substantive addition — how dynamical-stability constraints are imposed at *several lattice constants* rather than one. Every equation is stated in the form actually implemented, and every result is attributed to the script and output directory that produced it. Section 10 is the script/result map; Section 12 records four failure modes that were diagnosed along the way and which any write-up should mention, because three of them silently produced plausible-looking but wrong numbers.

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1 Scope and the one-sentence summary

A linear ACE potential is fitted by a quadratic program (QP) whose inequality rows encode physics: elastic (Born) stability, a pinned equilibrium lattice constant, and positive phonon frequencies throughout the Brillouin zone. A POPS committee is then drawn by rejection sampling under the same constraints. The finding this document exists to support is:

Imposing phonon stability at the equilibrium lattice constant *alone* is insufficient — constrained members are stable at a_{eq} yet leave the FCC lattice within picoseconds under NPT molecular dynamics. Imposing it on a grid of lattice constants spanning the thermal-expansion range fixes this, and yields a physically meaningful thermal-expansion coefficient.

2 Notation and the linearity that makes everything work

2.1 Linear ACE

For a configuration R of atomic positions, the ACE energy is *linear* in the parameter vector $\theta \in \mathbb{R}^K$ ($K = 91$ here):

$$E(R; \theta) = \sum_{k=1}^K \theta_k B_k(R) = \mathbf{B}(R) \cdot \theta, \quad (1)$$

where B_k are the ACE basis functions. This is the single structural fact the entire method rests on: because differentiation is linear, *every* derived quantity that is a linear or bilinear functional of E inherits an exact affine dependence on θ . In particular the force-constant (Hessian) matrix

$$\Phi_{a\alpha, b\beta}(\theta) = \frac{\partial^2 E}{\partial r_{a\alpha} \partial r_{b\beta}} = \sum_{k=1}^K \theta_k \underbrace{\frac{\partial^2 B_k}{\partial r_{a\alpha} \partial r_{b\beta}}}_{\Phi_{a\alpha, b\beta}^{(k)}} \quad (2)$$

splits into K geometry-only tensors $\Phi^{(k)}$. We call $\{\Phi^{(k)}\}$ the *undotted Hessian*: it depends on the geometry but not on θ , so it is built once per geometry and reused for every committee member. That is what makes a full-Brillouin-zone phonon check cheap enough to serve as a rejection-sampling predicate.

Implementation: `scripts/bandpath_phonon_uq/lib.jl`, functions `_site_basis_hessian` (nested ForwardDiff of `evaluate_basis` at site level) and `undotted_hessian` (threaded accumulation over atoms, with the four $\pm$ blocks per neighbour pair that enforce translational invariance).

2.2 Design system and preconditioner

Let $\mathbf{A} \in \mathbb{R}^{M \times K}$ be the design matrix (row m = basis evaluation for observation m), $\mathbf{y} \in \mathbb{R}^M$ the observations (energies/forces/virials), $\mathbf{W}$ the diagonal weight matrix, and $\mathbf{P}$ the diagonal preconditioner supplied by the fitting pipeline. Define

$$\tilde{\mathbf{A}} = \mathbf{W}\mathbf{A}\mathbf{P}^{-1}, \quad \tilde{\mathbf{y}} = \mathbf{W}\mathbf{y}, \quad \tilde{\boldsymbol{\theta}} = \mathbf{P}\boldsymbol{\theta}, \quad \lambda = 1/M. \quad (3)$$

All optimisation is carried out in the preconditioned variable $\tilde{\boldsymbol{\theta}}$ and mapped back by $\boldsymbol{\theta} = \mathbf{P}^{-1}\tilde{\boldsymbol{\theta}}$.

3 The constrained fit

3.1 Objective

In OSQP's convention $\min_{\mathbf{x}} \frac{1}{2}\mathbf{x}^\top \mathbf{P}_{\text{qp}}\mathbf{x} + \mathbf{q}^\top \mathbf{x}$ subject to $\mathbf{l} \leq \mathbf{A}_{\text{qp}}\mathbf{x} \leq \mathbf{u}$, the code sets

$$\mathbf{P}_{\text{qp}} = \tilde{\mathbf{A}}^\top \tilde{\mathbf{A}} + \lambda \mathbf{P}^\top \mathbf{P}, \quad \mathbf{q} = -\tilde{\mathbf{A}}^\top \tilde{\mathbf{y}}, \quad (4)$$

so that, up to an additive constant, the objective is

$$J(\tilde{\boldsymbol{\theta}}) = \frac{1}{2}\|\tilde{\mathbf{A}}\tilde{\boldsymbol{\theta}} - \tilde{\mathbf{y}}\|_2^2 + \frac{\lambda}{2}\|\mathbf{P}\tilde{\boldsymbol{\theta}}\|_2^2. \quad (5)$$

A subtlety worth stating explicitly in the paper. The regulariser is written in the *preconditioned* variable. Substituting $\tilde{\boldsymbol{\theta}} = \mathbf{P}\boldsymbol{\theta}$ gives $\frac{\lambda}{2}\|\mathbf{P}^2\boldsymbol{\theta}\|_2^2$, i.e. the effective Tikhonov prior in coefficient space is

$$\boldsymbol{\Gamma}_{\text{eff}} = \mathbf{P}^2, \quad (6)$$

not $\mathbf{P}$. This is a property of the implementation, not an accident of notation; if the paper reports the prior it should report $\mathbf{P}^2$.

3.2 Transporting constraint rows

A physical constraint is naturally written as a row $\mathbf{g}$ acting on the coefficients, $\mathbf{g} \cdot \boldsymbol{\theta} \geq c$. In the solver variable this becomes

$$\mathbf{g} \cdot \boldsymbol{\theta} = \mathbf{g} \mathbf{P}^{-1} \tilde{\boldsymbol{\theta}}, \quad (7)$$

which is why every constraint row appears in the code as `row/P` (right division by the diagonal $\mathbf{P}$) and the solution is recovered as `P\r . x`.

4 The POPS committee

4.1 Pointwise corrections (the “naive” forest)

Write

$$\mathbf{C} = \tilde{\mathbf{A}}^\top \tilde{\mathbf{A}} + \lambda \mathbf{P}^2, \quad \mathbf{G} = \mathbf{C}^{-1} \tilde{\mathbf{A}}^\top, \quad h_i = (\tilde{\mathbf{A}}\mathbf{G})_{ii}, \quad \mathbf{r} = \tilde{\mathbf{y}} - \tilde{\mathbf{A}}\tilde{\boldsymbol{\theta}}_{\text{LSQ}}, \quad (8)$$

where h_i is the leverage of observation i and r_i its residual. The POPS pointwise correction and the corresponding committee member are

$$\boxed{\delta\boldsymbol{\theta}^{(i)} = \mathbf{P}^{-1} \mathbf{G}_{:,i} \frac{r_i}{h_i}, \quad \boldsymbol{\theta}^{(i)} = \boldsymbol{\theta}_{\text{LSQ}} + \delta\boldsymbol{\theta}^{(i)}. \quad (9)}$$

Interpretation (this is the cleanest way to motivate POPS in the paper). The correction is exactly the perturbation that makes the model *interpolate* observation i . Indeed

$$\tilde{\mathbf{A}}_{i,:}(\mathbf{P} \delta \boldsymbol{\theta}^{(i)}) = \tilde{\mathbf{A}}_{i,:} \mathbf{G}_{:,i} \frac{r_i}{h_i} = h_i \frac{r_i}{h_i} = r_i, \quad (10)$$

so the member’s prediction at i moves by precisely the residual there, closing it. Each committee member is therefore “the fit that would have been obtained had observation i been reproduced exactly”; the ensemble spread measures how far the fit can be pushed by individual data.

Members are selected as the $n_{\text{lev}} = 5$ highest-leverage observations, the $n_{\text{res}} = 10$ largest- $|r|$ observations not already chosen, and $n_{\text{rand}} = 15$ random others, giving $N = 30$ members (Random.seed!(1234) makes this reproducible).

4.2 Constrained members

The constrained analogue replaces (9) by a QP that imposes the interpolation condition (10) as an *equality row* while adding the physics rows of Section 5:

$$\begin{aligned} \boldsymbol{\theta}_{\text{con}}^{(i)} = \mathbf{P}^{-1} \arg \min_{\tilde{\boldsymbol{\theta}}} \quad & J(\tilde{\boldsymbol{\theta}}) \\ \text{s.t.} \quad & \tilde{\mathbf{A}}_{i,:} \tilde{\boldsymbol{\theta}} = \tilde{y}_i \quad (\text{interpolate obs. } i) \\ & (\mathbf{b}' \mathbf{P}^{-1}) \tilde{\boldsymbol{\theta}} = 0 \quad (\text{equilibrium at } a_{\text{eq}}) \\ & l_{\text{phys}} \leq \mathbf{G}_{\text{phys}} \mathbf{P}^{-1} \tilde{\boldsymbol{\theta}} \leq \infty. \end{aligned} \quad (11)$$

The constrained *mean* model $\boldsymbol{\theta}_{\text{mean}}$ is the same program with the interpolation row removed.

5 The physical constraint rows

5.1 Equation of state: pinning the equilibrium

Let $\mathbf{b}(a) = \mathbf{B}$ (reference cell at lattice constant a), so that the per-cell energy is $E(a; \boldsymbol{\theta}) = \mathbf{b}(a) \cdot \boldsymbol{\theta}$ — again linear in $\boldsymbol{\theta}$. With $\mathbf{b}' = \text{d}\mathbf{b}/\text{d}a|_{a_{\text{eq}}}$ and $\mathbf{b}'' = \text{d}^2\mathbf{b}/\text{d}a^2|_{a_{\text{eq}}}$ (both by ForwardDiff):

$$\mathbf{b}' \cdot \boldsymbol{\theta} = 0 \quad (\text{equality}), \quad \mathbf{b}'' \cdot \boldsymbol{\theta} \geq \varepsilon, \quad \varepsilon = 10^{-9}. \quad (12)$$

The first states that a_{eq} is a stationary point of the energy–volume curve; the second that it is a minimum. Together they pin every committee member’s equilibrium lattice constant to the common reference a_{eq} , which is what makes a *fixed* reference geometry legitimate for the phonon rows.

5.2 Born elastic stability

The Voigt elastic operator is again linear in $\boldsymbol{\theta}$:

$$C_{ij}(\boldsymbol{\theta}) = \sum_k \theta_k H_{ij,k}^{\text{el}}, \quad H_{ij,k}^{\text{el}} = \left. \frac{\partial^2 B_k}{\partial \varepsilon_i \partial \varepsilon_j} \right|_{\boldsymbol{\varepsilon}=0}, \quad (13)$$

computed by `elastic_hessian_basis` in `src/Elasticity/get_hessian.jl`. The three cubic Born criteria are imposed as rows

$$C_{44} \geq 0.1, \quad C_{11} - C_{12} \geq 1.0, \quad C_{11} + 2C_{12} \geq 0.1, \quad (14)$$

in the raw (eV, per reference cell) units of H^{el} ; conversion to GPa uses $160.2176621/|\det \mathbf{L}_0|$ and is applied only for reporting. Stacking with (12) gives the matrix `born_rows` with lower bounds `born_lower=[0.1, 1.0, 0.1, 1e-9]`.

5.3 Phonons: the dynamical matrix is affine in θ

From the undotted Hessian (2), the Bloch transform to the primitive-cell dynamical matrix is a fixed linear map, so

$$\boxed{D(\mathbf{q}; \theta) = \sum_{k=1}^K \theta_k D^{(k)}(\mathbf{q})} \quad (15)$$

with, for primitive-cell atoms i, j and supercell atoms l ,

$$D_{i\alpha, j\beta}^{(k)}(\mathbf{q}) = \frac{1}{\sqrt{m_i m_j}} \sum_{l: s2p(l)=j} \Phi_{p2s(i)\alpha, l\beta}^{(k)} e^{i\mathbf{q} \cdot \mathbf{R}_l}, \quad \mathbf{R}_l = \text{MIC}(\mathbf{L}(\mathbf{f}_l - \mathbf{f}_j)), \quad (16)$$

where MIC is the minimum-image convention applied in *supercell* fractional coordinates (wrapping to $(-\frac{1}{2}, \frac{1}{2}]$), and D is Hermitised afterwards. Implementation: `precompute_force_constants` and `dynamical_matrix_from_fc` in `src/Phonons/phonon_bands.jl`; the per-basis assembly into $\{D^{(k)}\}$ over the whole band path is `bandpath_Dk` in `scripts/bandpath_phonon_uq/lib.jl`.

Frequencies follow from the eigenvalues μ_ν of D :

$$\omega_\nu(\mathbf{q}; \theta) = \text{sign}(\mu_\nu) \sqrt{|\mu_\nu|} \cdot f, \quad f = \frac{1}{2\pi \times 10^{12}} \sqrt{\frac{e}{\text{\AA}^2 u}} \approx 15.633 \text{ THz}, \quad (17)$$

the signed square root being the standard device for displaying imaginary (unstable) modes as negative frequencies.

Stability metric. Acoustic branches vanish at Γ , so the scalar used throughout is the minimum *non-acoustic* frequency over the sampled path:

$$m(\theta) = \min_{\mathbf{q}: \|\mathbf{q}\| \geq q_\Gamma} \min_{\nu} \omega_\nu(\mathbf{q}; \theta), \quad q_\Gamma = 5 \times 10^{-2}, \quad (18)$$

implemented as `min_freq_stable`. The path is $\Gamma \rightarrow X \rightarrow U \rightarrow L \rightarrow \Gamma \rightarrow K$ with per-segment sampling `N_per_seg=[20, 20, 20, 20, 60]` — the final $\Gamma \rightarrow K$ leg is sampled three times more densely because the soft modes concentrate there and a sharp dip between q -points would otherwise escape both the constraint and the check. Per-segment sampling required extending `fcc_band_path` in `src/Phonons/phonon_bands.jl` to accept a vector.

6 Cutting-plane repair

Requiring $\min_{\nu} \mu_\nu(\mathbf{q}; \theta) \geq \omega_{\text{cut}}^2$ is not a linear constraint on θ . It is, however, the infimum of linear constraints. For any fixed unit vector e , the Rayleigh quotient

$$e^\dagger D(\mathbf{q}; \theta) e = \sum_k \theta_k e^\dagger D^{(k)}(\mathbf{q}) e = g(\mathbf{q}, e) \cdot \theta, \quad g_k(\mathbf{q}, e) = \text{Re}[e^\dagger D^{(k)}(\mathbf{q}) e] \quad (19)$$

is an *exact linear functional* of θ — no finite differences, no displacement amplitude. This is `cut_row(iq, e, bp)`, implemented as `vec(ēe†)† Bq`, which reproduces $e^\dagger D^{(k)} e$ exactly under column-major `vec`.

Because $e^\dagger D e \geq \mu_{\min}(D)$ for every unit e , a single row (19) is *necessary but not sufficient*: enforcing it at the current softest eigenvector does not guarantee the new θ has $\mu_{\min} \geq \omega_{\text{cut}}^2$, because the eigenvector rotates as θ moves. The remedy is the standard cutting-plane loop: re-diagonalise, add rows for whatever is still soft, re-solve, iterate. Convergence is declared when no soft mode remains; `max_cuts=40` caps the iterations.

$$\omega_{\text{cut}}^2 = \left(\frac{\omega_{\text{cut}}^{\text{THz}}}{f} \right)^2, \quad \omega_{\text{cut}}^{\text{THz}} = 0.15 \text{ THz}. \quad (20)$$

Note the asymmetry between the two stages, which the paper should state: the *repair* stage imposes $\mathbf{b}' \cdot \boldsymbol{\theta} = 0$ exactly (an equality, $l = u = 0$), whereas the *rejection* stage cannot — an equality has zero acceptance probability under a continuous proposal — and instead imposes the band (24) below.

7 Hypercube proposal and rejection sampling

Collect the N constrained corrections as rows of $\boldsymbol{\Delta} \in \mathbb{R}^{N \times K}$, $\boldsymbol{\Delta}_{i,:} = \boldsymbol{\theta}_{\text{con}}^{(i)} - \boldsymbol{\theta}_{\text{mean}}$. Diagonalise the second-moment matrix and keep the numerically supported directions,

$$\boldsymbol{\Delta}^\top \boldsymbol{\Delta} = \mathbf{V} \boldsymbol{\Lambda} \mathbf{V}^\top, \quad \text{keep } j : \Lambda_{jj} > 10^{-8} \max_l \Lambda_{ll}, \quad (21)$$

project the corrections onto the retained eigenbasis, $\boldsymbol{\Pi} = \boldsymbol{\Delta} \mathbf{V}$, and take the axis-aligned bounding box

$$\ell_j = \min_i \Pi_{ij}, \quad u_j = \max_i \Pi_{ij} \quad (22)$$

(the `percentile_clipping=0` case of hypercube, i.e. the *unclipped* box: the feasible-set geometry is imposed exactly by rejection rather than approximated by shrinking the proposal). Proposals are

$$\mathbf{u} \sim \mathcal{U}[0, 1]^d, \quad \boldsymbol{\theta} = \boldsymbol{\theta}_{\text{mean}} + \mathbf{V}(\boldsymbol{\ell} + (\mathbf{u}_b - \boldsymbol{\ell}) \odot \mathbf{u}), \quad (23)$$

and are accepted iff all of the following hold (`rejection_sample_hypercube` with a predicate, `src/POPS/POPSRegression.jl`):

(i) Born rows (14) satisfied;

$$(ii) \quad \frac{|\mathbf{b}' \cdot (\boldsymbol{\theta} - \boldsymbol{\theta}_{\text{mean}})|}{\mathbf{b}'' \cdot \boldsymbol{\theta}_{\text{mean}}} \leq 0.1; \quad (24)$$

(iii) $m_v(\boldsymbol{\theta}) \geq \omega_{\text{cut}}^{\text{THz}} - 10^{-6}$ for every volume v .

Reading condition (24). To first order the shift in a member's equilibrium lattice constant is $\Delta a_{\text{eq}} \simeq -(\text{d}E/\text{d}a)/(\text{d}^2E/\text{d}a^2) = -\mathbf{b}' \cdot \boldsymbol{\theta} / \mathbf{b}'' \cdot \boldsymbol{\theta}$. Condition (24) is therefore, to leading order, the statement $|\Delta a_{\text{eq}}| \leq 0.1 \text{ \AA}$: a sampled member's equilibrium may drift by at most 0.1 Å from the reference. This is the continuous relaxation of the equality $\mathbf{b}' \cdot \boldsymbol{\theta} = 0$ used in the repair stage.

8 The multi-volume extension

Everything above is stated at a single reference geometry. The extension is to impose condition (iii) on a grid of lattice constants:

$$a_v = s_v a_{\text{eq}}, \quad s_v \in \{1.00, 1.02, 1.04, 1.06, 1.08, 1.10\}, \quad v = 1, \dots, 6. \quad (25)$$

Because $\Phi^{(k)}$ depends on geometry, each a_v requires its *own* undotted Hessian build and its own $\{\mathbf{D}^{(k,v)}(\mathbf{q})\}$; these are cached to disk as `results/undotted_Hbasis_4x4x4_a<a>.jls` (≈ 429 MB each at $4 \times 4 \times 4$, ≈ 1 – 2 min to build on 20 threads).

Three points of design, all of which the paper should state:

1. a_{eq} **remains the reference geometry.** The equation-of-state rows (12) and the Born rows (14) are still evaluated at a_{eq} only. The extra volumes add stability requirements; they do not move the equilibrium.
2. **The cutting plane pools volumes.** Each iteration collects soft modes from *all* v and appends one row (19) per triple $(v, \mathbf{q}, \nu)$. In practice this produced a median of 864 and a maximum of 1605 cut rows per member.

Algorithm 1 Volume-resolved constrained POPS committee

1. **Reference geometry.** Relax a_{eq} from the mean model. Build $\mathbf{b}', \mathbf{b}'', H^{\text{el}}$ at a_{eq} ; assemble (12), (14).
 2. **Volume grid.** For $v = 1..6$ with a_v from (25): build the undotted Hessian $\{\Phi^{(k,v)}\}$ and the band-path operators $\{\mathbf{D}^{(k,v)}(\mathbf{q})\}$. Cache to disk.
 3. **Constrained mean.** Solve (26) without the interpolation row; loop: collect soft modes over all v , append cut rows, re-solve, until none remain or `max_cuts`.
 4. **Proposal cloud.** Select $N = 30$ observations (leverage / residual / random). For each, solve (26) with its interpolation row and run the same multi-volume cutting-plane loop.
 5. **Proposal box.** Form Δ from the repaired members, diagonalise, take the unclipped bounding box (22).
 6. **Rejection sample.** Draw uniformly in the box; accept iff Born (14), band (24), and $m_v \geq \omega_{\text{cut}}$ for all v (softest first). Repeat to N members.
 7. **Verify and export.** Recompute m_v for every member at every volume; write the member $\times$ volume table.
-

3. **The predicate short-circuits, softest volume first.** The volumes are tested in ascending order of $m_v(\boldsymbol{\theta}_{\text{mean}})$, so most rejections are paid for at one volume rather than six.

The constrained fit for member i therefore becomes the program

$$\begin{aligned}
 \min_{\tilde{\boldsymbol{\theta}}} \quad & J(\tilde{\boldsymbol{\theta}}) \\
 \text{s.t.} \quad & \tilde{\mathbf{A}}_{i,:} \tilde{\boldsymbol{\theta}} = \tilde{y}_i, \quad \mathbf{b}' \mathbf{P}^{-1} \tilde{\boldsymbol{\theta}} = 0, \\
 & C_{44} \geq 0.1, \quad C_{11} - C_{12} \geq 1.0, \quad C_{11} + 2C_{12} \geq 0.1, \quad \mathbf{b}'' \cdot \boldsymbol{\theta} \geq \varepsilon, \\
 & g(\mathbf{q}, \mathbf{e}) \mathbf{P}^{-1} \tilde{\boldsymbol{\theta}} \geq \omega_{\text{cut}}^2 \quad \forall (v, \mathbf{q}, \mathbf{e}) \in \mathcal{S}^{(t)},
 \end{aligned} \tag{26}$$

where $\mathcal{S}^{(t)}$ is the accumulated cut set after t iterations.

9 NPT validation and the FCC-survival test

A committee member is validated by NPT molecular dynamics at zero pressure (Langevin thermostat + Monte-Carlo barostat, Molly.jl), $4 \times 4 \times 4$ FCC supercell (256 atoms), $T \in \{300, 500, 700, 900\}$ K, 10^4 equilibration + 2×10^4 production steps at $\Delta t = 1$ fs.

9.1 Lattice constant and expansion coefficient

With per-frame volumes V_f over production frames $\mathcal{P}$ and $N_{\text{sup}} = 4$,

$$a(T) = \frac{1}{|\mathcal{P}|} \sum_{f \in \mathcal{P}} \frac{V_f^{1/3}}{N_{\text{sup}}}, \quad \alpha = \left. \frac{1}{a_0} \frac{da}{dT} \right|_{\text{lin.fit}}. \tag{27}$$

Because the box volume changes under NPT, the RDF, MSD and coordination analyses all use each frame's *own* box for the minimum-image convention — using a fixed box is a silent error under a barostat.

9.2 The FCC-survival criterion

The essential guard. Define the mean coordination within a cutoff $r_c = 3.3 \text{ \AA}$,

$$\langle Z \rangle = \frac{1}{N} \sum_{i=1}^N \left| \{ j \neq i : r_{ij} < r_c \} \right|, \quad \text{FCC declared iff } \langle Z \rangle \geq 12 - \tau, \quad \tau = 1, \quad (28)$$

alongside the median nearest-neighbour distance, whose FCC value is $a/\sqrt{2}$. Perfect FCC gives $\langle Z \rangle = 12$; thermal disorder at 700 K gives 11.5; the transformed phase observed here gives ≈ 9.1 .

Why the *mean* and not the *maximum*. An earlier version of the analysis logged only $\max_i Z_i$, which remains pinned at 12 long after the lattice has transformed, and consequently reported “6.3% thermal expansion” for a cell that was no longer FCC. α in (27) is fitted *only* over temperatures satisfying (28); temperatures that fail are plotted but excluded.

10 Script and result map

All paths are relative to `/storage/astro2/phupfb/PhD/acestuff/ACEWorkflow`.

10.1 Shared machinery

Path	Provides
<code>scripts/bandpath_phonon_uq/lib.jl</code>	Undotted Hessian (<code>undotted_hessian</code> , <code>undotted_Hbasis</code> with disk cache), band-path operators (<code>bandpath_Dk</code>), ω and stability (<code>bands</code> , <code>min_freq_stable</code> , <code>soft_modes</code>), cutting-plane row (<code>cut_row</code>), committee band plots, test-set coverage.
<code>src/Phonons/phonon_bands.jl</code>	$D(\mathbf{q})$ from force constants (Eq. 16), <code>FREQ_THz</code> (Eq. 17), high-symmetry paths with per-segment sampling, <code>bulk_prim_super</code> .
<code>src/POPS/POPSRegression.jl</code>	corrections (Eq. 9), hypercube (Eq. 22), <code>rejection_sample_hypercube</code> (predicate form).
<code>src/Elasticity/get_hessian.jl</code>	<code>elastic_hessian_basis</code> (Born rows), <code>relax_lattice_constant</code> (a_{eq}).

10.2 Production runs: script $\rightarrow$ output directory

Output directories are under `models/Al_12_4_6A_2_/results/`.

Script	Output directory	Status / role
scripts/uq/bandpath_committee_undotted_Al_12_4_6A_2.jl	bandpath_undotted/	Original committee, $N_{\text{cell}} = 3$. Superseded — see Section 12(a).
scripts/uq/bandpath_committee_undotted_Al_12_4_6A_2_ncell4_densek.jl	bandpath_undotted_ncell4_densek/	Single-volume committee, $N_{\text{cell}} = 4$, dense $\Gamma \rightarrow K$. The “constrained at a_{eq} only” baseline.
scripts/uq/bandpath_committee_undotted_Al_12_4_6A_2_multivolume.jl	bandpath_undotted_multivolume/	The method of this document: six volumes, Eq. (25).
scripts/uq/npt_thermal_expansion_worst_member_Al_12_4_6A_2.jl	npt_thermal_expansion_naive_worst_member/, npt_thermal_expansion_constrained_worst_member/, npt_thermal_expansion_worst_member/	Single-volume NPT. One script, run three times with npt_vector, committee_subdir, N_cell_fc and outdir edited between runs — see the provenance note below.
scripts/uq/npt_multivolume_member_Al_12_4_6A_2.jl	npt_multivolume_softest/	The validation of this document: NPT on the multivolume committee, with the FCC-survival test (28).

Provenance note (important, and easy to get wrong). npt_thermal_expansion_worst_member_Al_12_4_6A_2.jl carries its configuration as top-of-file constants, so the three output directories above correspond to three *different* configurations of the same file, not three scripts. The header line of each thermal_expansion_summary.csv records which member was used, e.g. # npt_member=naive-worst ... con_soft_source=repaired[20]. Read that header before attributing any figure. In particular, the “constrained” curve in the earlier bands_before_after_constraining figures came from a *repaired* member, not a rejection sample, because the selection is arg min over the union repaired $\cup$ rejection.

10.3 Diagnostic and supporting scripts

Script	What it establishes
scripts/uq/diagnose_stale_cache_Al_12_4_6A_2.jl	Rebuilds the undotted Hessian cache from scratch and re-checks assembly ($\Delta \sim 7 \times 10^{-13}$). Ruled out cache staleness as the cause of the negative frequencies.
scripts/uq/bands_individual_hessian_naive_vs_constrained_Al_12_4_6A_2.jl	Band plots where <i>each</i> member gets its own from-scratch Hessian (22 Hessians), rather than the shared reference-geometry one. $\rightarrow$ bands_individual_hessian/.
scripts/uq/check_worst_naive_native4_Al_12_4_6A_2.jl	Draws naive hypercube samples and validates each with a native Hessian on its <i>own</i> relaxed $4 \times 4 \times 4$ cell; saves the first genuinely unstable θ . $\rightarrow$ npt_worst_naive_native4/.
scripts/uq/npt_worst_naive_native4_Al_12_4_6A_2.jl	NPT on that vector, with all phonon checks done natively.
scripts/bandpath_phonon_uq/replot_and_calibration.jl	Publication re-plots and committee calibration. $\rightarrow$ bandpath_phonon_uq/ (note: this run saved figures but <i>no</i> committee CSVs).

10.4 Output files of the multi-volume run

File	(in multivolume/)	bandpath_undotted_ Contents
committee_repaired.csv		$30 \times K$; the QP-repaired members, Eq. (26).
committee_rejection.csv		$30 \times K$; the rejection-sampled committee.
theta_mean.csv		the constrained mean model.
minomega_by_volume_naive.csv etc.		member $\times$ volume table of m_v , Eq. (18).
minomega_vs_volume.pdf		the stability-vs-volume figure.
bands_naive_v1.04.pdf etc.		band structures at each s_v , shared y -axis.
theta_npt_softest.csv, npt_candidates.csv		members handed to the NPT run.

11 Results

11.1 Committee stability across the volume grid

Minimum over the 30 members of m_v , in THz (bandpath_undotted_multivolume/minomega_by_volume_*.csv):

s_v	1.00	1.02	1.04	1.06	1.08	1.10
naive POPS	-8.656	-10.920	-11.471	-10.812	-9.043	-9.077
rejection (constrained)	+0.252	+0.175	+0.150	+0.166	+0.150	+0.153

Members unstable ($m_v < -0.05$) at *any* volume: constrained 0/30, naive 30/30. Rejection acceptance was 13.3% (226 proposals for 30 members); the binding volumes were $s = 1.10$ (131 rejections) and $s = 1.04$ (55), with $s = 1.06$ rejecting only 2. **The soft mode is not monotonic in volume** — constraining only the endpoints would have missed the $s = 1.04$ dip. Total wall time 21 minutes on 20 threads.

11.2 NPT validation

Member rejection[18] ($\min_v m_v = +0.150$ THz), from npt_multivolume_softest/thermal_expansion_summary.csv:

T (K)	a (Å)	$\Delta a/a_0$	$m(\theta)$ (THz)	$\langle Z \rangle$	med. NN (Å)	FCC?
0	4.0449	—	+0.288	12.00	2.860	yes
300	4.0779	+0.81%	+0.302	11.99	2.726	yes
500	4.0996	+1.35%	+0.290	11.85	2.689	yes
700	4.1204	+1.87%	+0.280	11.51	2.659	yes
900	4.3121	+6.61%	+0.208	9.12	2.498	no

Fitting (27) on the three FCC temperatures gives $\alpha = 26.3 \times 10^{-6} \text{ K}^{-1}$ with $R^2 = 0.99988$; the leave-one-out range is $[25.8, 26.8] \times 10^{-6}$. For comparison, aluminium’s experimental α is 23.1×10^{-6} at 300 K rising to $\approx 26 \times 10^{-6}$ by 700 K, so a 300–700 K fit should be compared against ≈ 26 , not against the room- temperature value.

Run	α (10^{-6} K^{-1})	leave-one-out	FCC temps
naive worst, single-volume	41.46	[34.4, 67.9]	0/4
constrained softest, single-volume	18.95	[14.8, 31.7]	0/4
constrained softest, six-volume	26.31	[25.8, 26.8]	3/4

The 900 K failure is of a different kind. At 900 K the lattice leaves FCC, but $a(900\text{ K}) = 4.3121\text{ \AA} = 1.066\text{ }a_{\text{eq}}$ lies *inside* the constrained grid (25), and the phonon spectrum there is *positive*, $m = +0.208\text{ THz}$. The transition therefore occurs at a geometry where the model is harmonically stable: it is a finite-amplitude, thermally activated transition to a competing minimum, not a dynamical instability. Harmonic constraints of the form (19) cannot address it by construction. (900 K is also 96% of aluminium’s melting point, 933 K.)

The natural next constraint. Because E is linear in θ (1), the requirement that FCC lie below a specific competing structure R_c is exactly *one* linear row,

$$(B(R_c) - B(R_{\text{FCC}})) \cdot \theta > 0, \quad (29)$$

per atom, requiring only two basis evaluations and no Hessian. A relaxed frame of the competing phase is available in `npt_multivolume_softest/T900K/md_trajectory.extxyz`. This is untested.

12 Four pitfalls that produced plausible but wrong numbers

Each of these was diagnosed during development and each, while active, produced output that looked reasonable. They are worth a methods paragraph.

(a) Supercell too small for the cutoff. With a $6\text{ \AA}$ model cutoff, a $3 \times 3 \times 3$ FCC supercell has box half-width $\approx 6.07\text{ \AA} \approx$ the cutoff, so an atom interacts with its own periodic image and zone-boundary phonons are aliased. Converged values require $N_{\text{cell}} \geq 4$. The effect is large and not a small correction: one member gave $+0.150\text{ THz}$ at $N_{\text{cell}} = 3$ and -5.199 THz at $N_{\text{cell}} = 4$ and 5 . *All* phonon numbers computed at $N_{\text{cell}} = 3$ are unreliable, including those in `results/bandpath_undotted/`. Diagnostic: `scripts/uq/phonon_pipeline_ncell_diagnosis_A1_12_4_6A_2.jl`. A cheap guard is to assert $N_{\text{cell}} a/2 > \text{cutoff}$.

(b) Fixed-geometry checks applied to unpinned vectors. The band-path operators $\{D^{(k)}\}$ are built at one geometry. This is valid for constrained members precisely because $\mathbf{b}' \cdot \theta = 0$ pins their equilibrium there — but *not* for naive POPS vectors, whose own relaxed a differs. Naive members must be checked with a Hessian built at their own relaxed lattice constant (`scripts/uq/check_worst_naive_native4_A1_12_4_6A_2.jl`).

(c) Maximum coordination as a crystallinity test. $\max_i Z_i$ stays at 12 in a transformed cell; only $\langle Z \rangle$ falls. Using the maximum caused a transformation ($\langle Z \rangle : 12 \rightarrow 9.5$, NN $2.86 \rightarrow 2.58\text{ \AA}$, $\Delta E \approx -0.18\text{ eV/atom}$) to be reported as thermal expansion. See Eq. (28).

(d) Fitting α across a phase change. Because α is a slope, a constant offset from a transformation cancels out of it, so a fit over transformed points can look plausible. The single-volume run’s $\alpha = 18.95 \times 10^{-6}$ was of this kind: superficially close to experiment, but fitted entirely on non-FCC data and unstable to refitting (leave-one-out [14.8, 31.7]). Quote leave-one-out ranges, and fit only FCC points.

13 Reproducing

```
cd /storage/astro2/phupfb/PhD/acestuff/ACEWorkflow
JID=$(sbatch --parsable scripts/uq/run_committee_multivolume.slurm)
sbatch --dependency=afterok:$JID scripts/uq/run_npt_multivolume.slurm
```

Both are `hmem`, 20 tasks. Observed wall times: committee 21 min, NPT 4 h 27 min. The committee job builds five new ≈ 429 MB Hessian caches (a_{eq} is reused from the single-volume run). Determinism comes from `Random.seed!(1234)` in both scripts.

Two open items. (i) Test-set coverage for the multi-volume committee was 48.1% (energy RMSE 0.972 eV); the corresponding single-volume figure was not retained, so whether the tighter constraints cost calibration is *not yet established*. (ii) The committee sits on the constraint boundary — the softest member has $\min_v m_v = +0.1503$ THz against $\omega_{\text{cut}}^{\text{THz}} = 0.15$, and the median member $+0.1586$ — so `:softest` and `:median` are effectively the same vector. Members with genuine headroom require a larger ω_{cut} .